

Editorial

Beyond the prescription: Addressing socio-clinical determinants of asthma care in India

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Received April 10, 2026; Accepted April 15, 2026; Published May 16, 2026

Asthma is a common chronic airway disease and is the second leading cause of mortality among the chronic respiratory diseases.¹ This disease affects about 300 million people globally and leads to about 1000 deaths per day.² Although there has been a global decrease in incidence, prevalence, mortality rates, and disability-adjusted-life-years (DALYs), the absolute numbers for incidence, prevalence, and DALYs remain high, especially in areas with a low SDI.^{3,4} Forecasting models predict that the increases in the total number of asthma will probably continue until 2050, even as age-standardised prevalence rates are expected to remain stable.⁴ India contributes to about 13% of the global burden with an estimated 35 million cases.⁵ However, the rates of mortality and DALYs due to asthma in India are threefold and twofold higher than the global rates.⁶ The disease exacts a whopping economic burden, conservative estimates putting the cost of treatment at more than 487 billion INR.⁷

Despite advances in management of asthma in recent decades, most patients in India remain inadequately managed. Underdiagnosis is frequent due to the non-availability or under-use of diagnostic tools such as spirometry or even peak flowmetry.⁶ A recent study reported that 82% of patients with early symptoms and 70% with severe asthma were clinically underdiagnosed in India, with suboptimal utilisation of pulmonary function testing.⁵ Similarly, management remains suboptimal in most peripheral settings, with the majority of patients poorly controlled.⁸

Access to proper asthma care and its affordability remain major roadblocks for so many patients. These

continue to plague patient management despite huge advances in newer drugs, including the newer biologics, better drug delivery devices, and clearer treatment guidelines. The primary goals of management, viz., control of symptoms and prevention of exacerbations, remain elusive. Thus, despite filled prescriptions, outcomes lag. While physicians fixate on the biological pathophysiology of airway inflammation and reversible obstruction, concurrent socio-clinical determinants get little attention. The providers fail to recognise that interwoven with the patient factors are social, economic, environmental, and systemic factors that dictate access, adherence, and equity in asthma care.⁹ Urban-rural divide exacerbates disparities; while metro cities boast specialist clinics and tertiary care settings, rural patients, especially in lower sociodemographic index (SDI) States, rely on health centres with inadequate personnel, diagnostic facilities, and drug supplies. Socioeconomic pressures compound this, with a large proportion of patients facing out-of-pocket expenses, which is catastrophic for daily wagers. Gender biases persist with women experiencing delays in care due to several social, economic mobility and domestic constraints.¹⁰ Consider a young farmer from rural Bihar presenting with nocturnal wheeze and productive cough. His peak expiratory flow oscillates wildly, signalling poor control. Inhaler technique is found to be faulty; he puffs it like a cigarette, delivering a measly fraction of the dose; no one ever taught him the inhaler usage. Add to this the biomass *chulha* at home, dusty roads, and moulds and other irritants at his workplace. This is a perfect blend of socio-clinical

This editorial is published on the occasion of World Asthma Day-May 5, 2026

How to cite this article: Koul PA, Saydain G. Beyond the prescription: Addressing socio-clinical determinants of asthma care in India. *Indian J Med Res.* 2026;163:565-8. doi: 10.25259/IJMR_1285_2026

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profile of poverty, pollution, poor education, overriding the pharmacological therapy that is generally focused upon by the physicians. Recognising such scenarios, the current guidelines mandate social prescribing, urging assessment of determinants before stepping up therapy.²

The Asthma Insights and Reality in India¹¹ study showed that only 36% use controller medications, often linked to cost. Several misconceptions and misperceptions exist regarding the use of inhalational medications, such as ‘once started, a patient remains addicted to inhalers’. Social stigmas for the use of inhalers abound, which contributes to hesitancy.¹¹

India is a large landmass country, also being the most populous, with a heterogeneous geographic, cultural, and socio-economic profile. Previous studies have implicated several environmental triggers for symptoms of asthma that have included smoking,¹² truck traffic near the house,¹² open fire cooking,⁵ dampness in homes,¹³ obesity,¹⁴ paracetamol¹⁵ and antibiotic use.¹⁶ Breastfeeding¹⁷ and eating plenty of fresh fruits and vegetables can offer protection. In the recently published Global Asthma Network (GAN)-phase 1 study¹⁸ which included 127,000 participants across nine diverse centres, with a prevalence of current wheeze ranging from 3.16 to 3.63% across age groups, the data underscore a persistent challenge that necessitates evidence-based management and targeted policy interventions. In this study, the presence of allergic rhinitis, eczema, hay-fever, exposure to farm animals and chest infections in the first year of life, damp walls at home, and use of mosquito coils were associated with a higher prevalence of asthma symptoms.¹⁸ A parental history of asthma remained one of the strongest predictors for current wheeze and severe asthma across all age groups, with an adjusted odds ratio (AOR) as high as 2.88 in adults. An important aspect of the GAN study was to look for India-specific triggers, demonstrating that the consumption of curd (AOR 1.49) and ice cream (AOR 1.31) was associated with current wheeze in adolescents. Further, the use of mosquito coils and incense sticks was identified as a notable trigger (AOR 1.11), suggesting that indoor chemical irritants are as relevant as traditional allergens. Interestingly, caesarean section delivery emerged as a risk factor for adolescents (AOR 1.28), aligning with global discussions on the role of the early microbiome in immune development.

The Indian landscape presents unique environmental challenges. The GAN study findings bring renewed

focus to indoor air quality and its impact on the most vulnerable groups. Environmental determinants loom largest in India's asthma epidemic. Air pollution, with PM_{2.5} levels exceeding World Health Organization (WHO) limits is recorded in about 80% of Indian cities,¹⁹ and is noted to trigger a 21-29% increase in emergency room visits of asthmatic children on high and moderate pollution cluster level days compared to low pollution cluster days.^{20,21} Spikes of poor AQI correlate with ER visits in Delhi; similarly, stubble burning in Punjab-Haryana funnels asthmatics into overflow wards.^{20,22} Climate change amplifies this; erratic monsoons foster mould, while heatwaves provoke exercise-induced symptoms. In coastal Andhra Pradesh, rising humidity drives fungal sensitisation, yet allergy testing remains a luxury.²³ The use of coal, kerosene, or cow dung for cooking remains a major risk factor for adults, increasing the risk of current wheeze and severe asthma by 1.48 times.¹⁸ Moisture or damp spots in the side houses were consistently associated with wheezing in both children and adults, likely due to fungal growth.¹⁸ For adolescents, frequent truck traffic in front of the house significantly correlated with asthma symptoms, highlighting the toll of vehicular pollution.¹⁸ Cultural practices, traditional beliefs, and dietary habits also play a role in shaping perceptions regarding respiratory illnesses.

Healthcare system gaps perpetuate inequity. India's asthma specialist density is abysmal, 0.5 per million versus 5-10 in high-income countries. Many healthcare providers, both in private and government sectors, lack the necessary knowledge and skills to diagnose and treat asthma, often resorting to prescribing only oral drugs. Pharmacotherapies recommended for advanced stages or severe asthma like omalizumab or mepolizumab remain for the elite. PHCs stock salbutamol but skimp on ICS-formoterol combos, forcing reliever-only reliance, a recipe for crises. On top of everything, funding for asthma care remains woefully inadequate in a country that already spends only a small proportion of its GDP on healthcare. The allocation of adequate funds to asthma care in a country with poor GDP spending on healthcare,⁷ remain a concern even though the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases & Stroke (NPCDCS) nods to asthma but lacks dedicated funding.

To transcend prescriptions, we must integrate socio-clinical strategies. First, empower primary care with ‘Asthma Action Teams’: nurses or community health workers trained in inhaler demos, peak-expiratory flow (PEF) monitoring, inhaler use and trigger

counselling. Digital innovation in the current era of its high penetration helps. Custom asthma trackers with local vernacular voice prompts could boost self-management, especially for illiterate patients. India's young population, with 65% of people under the age of 35,²⁴ provides a real opportunity and advantage.

Policy overhaul is imperative. Universal inhaler coverage under *Ayushman Bharat*, currently patchy, must prioritise ICS/LABA fixed-dose combos, rather than short-acting beta agonists. Global Initiative for Asthma (GINA) has set a powerful and provocative theme for World Asthma Day 2026: 'Access to anti-inflammatory inhalers for everyone with asthma – still an urgent need'.²⁵ This theme is more than just a slogan; it is a clinical and ethical imperative. Despite decades of evidence showing that asthma is primarily a disease of airway inflammation, millions of patients globally still rely on short-acting beta2-agonists alone. This reliance is not just outdated but is outright dangerous. The policy planners need to consider subsidising anti-inflammatory medications *via* public platforms like the *Jan Aushadhi* for cases of asthma. Critics argue that resources are scarce, but cost-effectiveness speaks: every 1 USD invested in adherence, on average, yields 2.72 USD in return on investment.²⁶

Beyond the prescriptions lies a holistic ecosystem. Physicians need to advocate for cleaner energy, stricter air quality regulation, public education, and equitable access to agents that control asthma. By addressing these multifaceted risk factors, we can move closer to a future where every Indian can breathe freely. India's asthma story isn't written in spirometry curves alone—it is etched in equity, access, and action.

Financial support and sponsorship: None.

Conflicts of Interest: None.

Use of Artificial Intelligence (AI)-Assisted Technology for manuscript preparation: The authors confirm that there was no use of AI-assisted technology for assisting in the writing of the manuscript and no images were manipulated using AI.

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